

**IMPROVING MALAYSIA'S RICE SELF-SUFFICIENCY LEVEL WITH
A SYSTEM DYNAMICS APPROACH**

ISAAC YANG HAO TUNG

BACHELOR OF SCIENCE (DECISION SCIENCE)

UNIVERSITI UTARA MALAYSIA

2026

ABSTRACT

Rice is the staple food of Malaysia and plays a vital role in ensuring the country's food security. However, domestic rice production remains insufficient to meet national demand, resulting in continued dependence on imported rice and raising concerns about Malaysia's rice self-sufficiency level (SSL). This study aims to develop a System Dynamics (SD) model to analyse the factors affecting rice production and evaluate policy interventions that can improve local rice production and the national rice SSL. The model was developed using secondary data collected from government reports, journal articles, and published literature, and was validated through behaviour and structure verification. A base-case simulation was first conducted to represent the current state of the rice production system, followed by several intervention scenarios to assess their long-term impacts. The simulation results indicate that, without intervention, the gap between rice production and demand continues to widen over time. Among the intervention scenarios, increasing people per hectare contributed to bigger improvements in rice production and the rice self-sufficiency level. The findings demonstrate that System Dynamics is an effective decision-support tool for understanding the complex interactions within Malaysia's rice industry and for evaluating policies aimed at strengthening the country's long-term food security.

Keywords: Food security, Self-sufficiency, rice, System Dynamics

ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to God for His countless blessings, guidance, strength, and mercy throughout this research journey. Completing this undergraduate research is something I once thought was beyond my capabilities. Looking back, I realize that every challenge, every moment of doubt, and every achievement along the way was only possible through His grace. This accomplishment marks not only the completion of my degree but also the beginning of many future aspirations.

I would like to extend my sincere appreciation to my research supervisor, Assoc. Prof. Dr. Jafri Hj Zulkepli Hew for the invaluable guidance, patience, encouragement, and unwavering support provided throughout this study. Your willingness to share your knowledge, provide constructive feedback, and dedicate your time to every consultation has been instrumental in the successful completion of this research. Your commitment has inspired me to persevere, even during the most challenging moments.

I am deeply grateful to my parents and family for their unconditional love, prayers, sacrifices, and continuous encouragement. Their emotional and moral support has been my greatest source of strength, motivating me to keep moving forward despite every obstacle.

My heartfelt thanks also go to my friends for their companionship, encouragement, and shared experiences throughout this journey. The countless discussions, late nights, and mutual support made this demanding process much more meaningful and manageable.

Finally, I would like to thank myself for remaining resilient throughout this journey. Thank you for choosing perseverance over giving up, for believing that every hardship would eventually lead to growth, and for continuing to move forward even when the path seemed overwhelming. This achievement is a reminder that determination, faith, and perseverance can overcome even the greatest challenges.

May this accomplishment serve as the foundation for greater success in the future, and may everyone who has contributed to this journey be blessed abundantly

TABLE OF CONTENTS

ABSTRACT	ii
ACKNOWLEDGEMENT	iii
CHAPTER 1: INTRODUCTION	1
1.1 Background of Study	1
1.2 Problem Statement.....	2
1.3 Research Questions	3
1.4 Research Objectives.....	3
1.5 Scope of Research	4
1.6 Significance of Research	5
CHAPTER 2: LITERATURE REVIEW	6
2.1 Problems in Paddy and Rice	6
2.2 Breakdown of The Rice Industry in Malaysia.....	7
2.3 Problems faced by paddy and rice production in Malaysia	7
2.4 Techniques Used to Solve the Rice Economy	8
CHAPTER 3: METHODOLOGY	10
3.1 Introduction.....	10
3.2 Data Source and Collection.....	10
3.2.1 Secondary Data	10
3.3 Research Process.....	10
3.3.1 Problem Articulation	10
3.3.2 Formulation of Dynamic Hypothesis.....	11
3.3.3 Formulation of Simulation Model	11
3.4 Summary of Chapter	12
CHAPTER 4: ANALYSIS AND FINDING	13
4.1 Model Testing.....	13
4.1.1 Structure Validation Test.....	13
4.1.2 Behaviour Validation Test.....	13
4.2 Policy Formulation and Evaluation.....	15
4.2.1 Base Case Result	15
4.2.2 Intervention 1 — Increase Person Per Hectare to 300,000	16
4.2.3 Intervention 2 — Increase Subsidy to 60%	17
4.2.4 Intervention Result	19
4.2.5 Possible Drawbacks from Intervention.....	19
4.3 Summary.....	19

CHAPTER 5: CONCLUSION.....21

5.1 Conclusion21

5.2 Limitations.....21

LIST OF REFERENCES23

APPENDIX.....25

CHAPTER 1:

INTRODUCTION

1.1 Background of Study

Rice is an important food in Malaysia. The Malaysian population consumes more than 2.4 million tonnes of rice and rice products on a yearly basis. However, Malaysia's local rice industry is only able to meet 75% of the demand while the rest of the demand is met by import (Ahmad Zairy, 2023).

In 2020, then Agriculture and Food Industries Minister, Datuk Seri Dr Ronald Klandee, highlighted that Malaysia's self sufficiency level target for rice will be increased to 75% for the year 2025. But experts have said that this aspiration is unlikely to be achieved in a recent report (Annuar & Hisham, 2025)

The gap between domestic rice production and national demand highlights the complexity of Malaysia's rice industry. Various interconnected factors, such as paddy cultivation area, productivity, population growth, consumption patterns, and imports, collectively influence the country's self-sufficiency level. Understanding how these factors interact over time is essential for developing effective policies to improve rice production and reduce dependence on imports.

System Dynamics attempts to understand complex systems by replicating the way a system actually operates and it focuses on explanation or policy making (Sharp, 1977). This approach will allow us to explore the cause-and-effect relationships in the rice industry, test policy implementations, and simulate long-term behaviour.

1.2 Problem Statement

Despite the government providing various incentives, subsidies, and support programmes, the performance of Malaysia's local rice production continues to fall short of expectations. This gap between policy intentions and actual outcomes suggests that the challenges facing the rice industry extend beyond individual issues and are rooted in complex structural problems.

A proposed Seed Quality Bill has come under fire for potentially driving up costs for farmers who rely on uncertified seeds (Annuar & Hisham, 2025). Climate change has also posed a big threat to the local rice production by affecting soil fertility, temperature, and weather variables (Abdullah, 2025).

In addition to economic and environmental challenges, sociological factors also hinder improvements in rice productivity. The limited implementation of Good Agricultural Practices (GAP), low participation of farmers in training programmes, and the inability of some trainers to effectively transfer agricultural knowledge have significantly reduced productivity despite the government's continued support (Ahmad Zairy, 2023).

These issues do not exist independently but interact with one another, creating feedback effects that influence the overall performance of Malaysia's rice industry. As a result, addressing individual factors in isolation may produce limited or even unintended outcomes. A holistic and systematic approach is therefore needed to understand these complex relationships and evaluate the long-term consequences of different policy interventions.

System Dynamics provides such an approach by modelling the interactions and feedback mechanisms within complex systems. Through simulation, it enables policymakers to examine the potential impacts of various interventions before implementation, leading to more informed and effective decision-making. Therefore, this study employs System Dynamics to analyse the interactions among the key factors affecting Malaysia's rice industry and to evaluate policies aimed at improving local rice production and the national rice self-sufficiency level.

1.3 Research Questions

1. What are the key factors influencing Malaysia's rice production and how are the factors interconnected?
2. How do the key components and feedback relationships within Malaysia's rice production system influence rice production and self-sufficiency over time?
3. How do existing and alternative policy interventions affect the long-term behaviour of Malaysia's rice production system and self-sufficiency?

1.4 Research Objectives

1. To identify the key factors influencing Malaysia's rice production system and examine the interconnections among these factors.
2. To develop a System Dynamics model that captures the key components and feedback relationships affecting rice production and
3. To evaluate the long-term effects of existing and alternative policy interventions on Malaysia's rice production system and rice self-sufficiency.

1.5 Scope of Research

1. Case Study Location

This study is limited to Malaysia at a national-level. This means while state level rice production may differ, this model operates on a macro-level, capturing overall dynamics nationally.

2. Simulation Tool

The simulation model will be developed and analyzed based on the Vensim Simulation Software which is based on the System Dynamics and enables the analysis of the behaviour of dynamic systems in the long-term.

3. Boundary

This research is limited to Malaysia's rice production system, focusing on the internal structure and feedback mechanisms that influence rice output and self-sufficiency. External factors such as international political shocks, global climate change at a macro scale, and detailed consumer dietary preferences are excluded, except where they indirectly influence the system through policy or production-related variables.

1.6 Significance of Research

This research provides both theoretical and practical contributions to the understanding of Malaysia's rice production system and its self-sufficiency challenges. By applying a System Dynamics approach, the study offers insights into the complex interactions and feedback mechanisms that influence rice production over time.

From an academic perspective, this study contributes as literature on agriculture system modelling and food security by demonstrating how System Dynamics can be used to analyse rice production systems in a developing country context. Unlike statistical methods, System Dynamics captures dynamic behaviour and non-linear relationships within the system, thereby enhancing our understanding of how policy interventions and structural changes affect rice production and self-sufficiency in the long term.

The findings of this research may assist policymakers in identifying key leverage points within the rice production system. By simulating different policy scenarios, the model allows decision-makers to evaluate the potential long-term impacts of interventions. This can support more informed and sustainable policy design aimed at improving national rice self-sufficiency.

Additionally, this study benefits farmers and agricultural planners by highlighting systemic constraints and reinforcing mechanisms that affect production outcomes. Understanding these dynamics may help stakeholders better anticipate unintended consequences of policies and avoid short-term solutions that undermine long-term system performance.

Overall, this research contributes to evidence-based decision-making in Malaysia's agricultural sector and supports national efforts to realise its rice self-sufficiency target. By providing a structured, system-oriented framework, the study offers a foundation for future research and model extensions in agricultural planning and policy analysis.

CHAPTER 2:

LITERATURE REVIEW

2.1 Problems in Paddy and Rice

Rice represents a large portion of the world's diet, with 4 billion people being fed daily and even representing 70% of an individual's calorie intake in some countries. Tackling issues in the rice and paddy industry requires a true understanding of the cost of rice cultivation and connecting the dots between a diverse range of problems (Mills, 2020). One of the issues that requires solving is the problem of climate change. Agricultural productivity is affected by both direct and indirect consequences of climate change, including shifting rainfall patterns, droughts, floods, and the changing geographic spread of pests and diseases. Research indicates that over the last half-century, human-driven climate change has slowed the growth of agricultural production, particularly in mid- and low-latitude regions (Poertner, 2023).

However, rice is not just affected by climate change but also plays a part in affecting climate change. Research reveals that since the 1960s, global greenhouse gas emissions from rice cultivation have almost doubled, reaching approximately 1.1 billion tons of CO₂-equivalent annually during the 2010s—a figure comparable to the yearly output of 239 million vehicles. One major contributor to the emissions is the fertilizer, with the use of synthetic nitrogen increasing powerful greenhouse gases (Tian, 2026).

Aside from environmental factors, paddy and crop productivity also have to go up against sociological factors. Infrastructure plays a major part in improving crop productivity. A study in Luang Prabang Province, Laos, found that poor road infrastructures that connect farmers' homes to the rice fields provided a major constraint on rice production (Khamko Thammavong, 2020). The same study also found that the knowledge of farmers played a major part in rice production, contributing to an increase in adoption of technology that increased rice production. When farmers have a firm grasp of agronomic knowledge, the more likely they are to manage their fields effectively (Sokra, 2026).

Economic factors also play a major role in rice production. A study found that the lack of support from financial agencies led to a lower rice production as this meant not investing in farming equipment such as power tillers (Khamko Thammavong, 2020). Manual labour is critical to rice cultivation; consequently, worker shortages during key phases like transplanting and harvesting can delay vital operations and diminish overall crop yields. But manual labour

comes with salaries to pay, which are increasing by the minute in many regions. This forces farmers to adopt labour-saving practices which decreases rice yields (Sokra, 2026).

2.2 Breakdown of The Rice Industry in Malaysia

Historically, Malaysia maintained an agriculture-centric economy, characterized by the robust production of diverse commodities including rice, palm oil, and cocoa (Zainudin, 2025). After Malaysia's independence in 1957, the Malaysian Agricultural Research and Development Institute (MARDI) was established to lead research on agriculture which included rice and paddy (Dorairaj, 2023).

In 2023, Kedah and Perlis represented 44% of the total paddy area and 45.6% of the total paddy production. In the same year, Sekinchan, Selangor, achieved the highest rice yield in Malaysia with a production level of 9,000 to 12,000kg per hectare. Even with that, the self-sufficiency level of rice in Malaysia has had a steady decline from 70% to 56.2% (Suruhanjaya Persaingan Malaysia, 2024). Current Agriculture and Food Security Minister Datuk Seri Mohamad Sabu has stated that the ratio is low due to underutilising paddy areas and the slow development of new granary areas (Khoo, 2026).

Malaysia's rice self-sufficiency remains a significant policy concern due to the country's continued dependence on imported rice. Existing literature suggests that rice production is influenced by a complex interaction of environmental, economic, infrastructural, and sociological factors.

2.3 Problems faced by paddy and rice production in Malaysia

Climate change has been consistently identified as one of the most important threats to rice production. (Firdaus, 2020) found that changes in temperature and precipitation negatively affect rice yields and threaten food security in Malaysia. Similarly, (Abdullah, 2025) demonstrated that extreme weather events such as floods, driven by changing precipitation patterns, reduce paddy productivity in granary areas. (Solaymani, 2023) further reported that increases in temperature can significantly reduce rice production in the long run, while (Tan, 2021) highlighted that temperature effects differ between planting seasons, indicating the complexity of climate impacts on rice yields.

Beyond environmental factors, several studies emphasize the importance of human and institutional factors. (Ahmad Zairy, 2023) found that increased support from agricultural extension officers positively influences rice yields through improved knowledge transfer and farmer engagement. Likewise, (Baharudin, 2021) stressed the importance of collaboration between farmers, researchers, and government agencies to improve agricultural outcomes. (Dorairaj, 2023) further noted that poor technology adoption, limited farmer awareness, and weak implementation of agricultural best practices continue to constrain productivity.

Infrastructure and resource availability also play a crucial role in rice production. (Mgale, 2020) argued that improved rural infrastructure enables farmers to access information, markets, and resources more effectively, thereby improving productivity. This finding is supported by (Abidin et al, 2022), who identified rural infrastructure, labour, and capital investment as significant contributors to rice production performance in Malaysia.

Land availability represents another important challenge. (Fróna, 2020) observed that changing food consumption patterns can alter land-use priorities and reduce the availability of arable land for crop production. In the Malaysian context, (Suffian & Suffian, 2023) highlighted competition between rice cultivation, oil palm plantations, and development projects as a major constraint on agricultural land availability.

2.4 Techniques Used to Solve the Rice Economy

(Solaymani, 2023) implements a method known as the Breitung–Candelon Spectral Causality and it found that a 2% increase in temperature reduces rice production by 9% in the long run and vice versa. (Tan et al., 2025) uses Regression Modelling to examine the impact of climate change on rice yield. The Markov Chain Monte Carlo method was used to estimate the impact of these climate factors on paddy yields (Abdullah, 2025), which found that precipitation at different times can affect rice yields by introducing extreme weather occurrences such as floods. The Mann–Kendall and Sen’s slope test was used in this study (Firdaus, 2020) to examine the trends of maximum and minimum temperatures and precipitation, and the same study found that Climate Change poses a huge threat to granary areas with its effects affecting temperature and precipitation, which would lead to a decrease in rice yields.

Although previous studies have successfully identified individual factors affecting rice production, most have examined these factors in isolation using statistical or econometric

methods. Limited research has explored how these factors interact through feedback relationships over time.

Several studies have carried out comprehensive analysis on food security with the use of System Dynamics. One study (Çelik & Serin, 2022) utilised System Dynamics to examine food security in Turkey and its factors, the study found that optimal land use and reduction of food wastes will positively impact the future food security of Turkey. Another study (Eneh & Eneh, 2025) also used System Dynamics to analyse agricultural practices and food security in Nigeria, this study found that agricultural productivity alone does not guarantee food security. Several underlying factors, such as the effectiveness of food distribution systems, economic access to food, and regional disparities, also play significant roles to sustain food security.

Although these studies use System Dynamics to simulate long-term behaviour in food security systems, the scope of our study is minimised to only the rice and paddy industry in Malaysia. Whereas food security represents the broader range of products for consumption.

Therefore, this study adopts a System Dynamics approach to examine the interconnected nature of Malaysia's rice production system and evaluate the long-term effects of alternative policy interventions on rice self-sufficiency.

CHAPTER 3:

METHODOLOGY

3.1 Introduction

In this chapter, the methodology of this study will be explained in more detail. It includes the data source and collection, research process, expected result and Gantt Chart. Methodology in research explains how a researcher plans a study systematically to ensure that the results are valid and trustworthy and address the research's goals and objectives. System Dynamic Modelling (SDM) is the technique that will be used in this study. Since the development of this model uses SDM techniques, it must follow the SDM process. That is, there are five processes. It will be described in more detail in the process section. Without these processes, the model to be developed will not be able to as what is desired next will not be able to achieve the primary purpose of this study.

3.2 Data Source and Collection

3.2.1 Secondary Data

Data obtained earlier than primary data are referred to as secondary data. The data will be collected using the appropriate source data to develop the model. Data taken are from the (Statista, 2026). Variables related to the model's development will be taken and used to develop the Causal Loop Diagram in Formulation of Dynamic Hypothesis.

3.3 Research Process

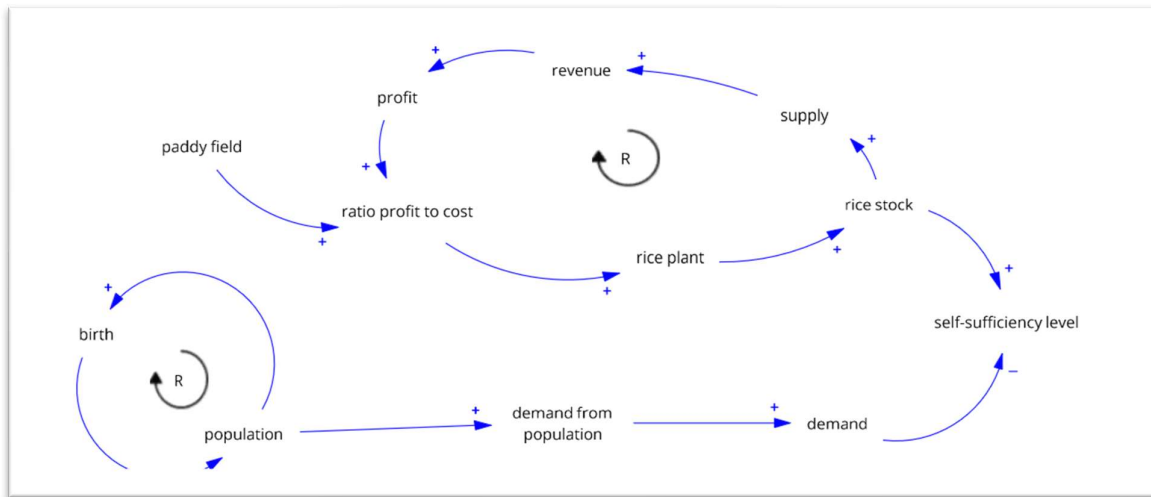
Each technique used in the study has its process. Without a process, the result cannot be produced. SDM has five process modelling. Besides, modelling is a feedback process, not a linear sequence of steps. Models go through constant iteration, continual questioning, testing, and refinement. Therefore, modelling is an iterative process. Iteration can occur from any step to any other step.

3.3.1 Problem Articulation

The main topic for this study is the self-sufficiency level of rice in Malaysia. Rice stock in this country cannot fulfil the total demand for rice because of increase in population. It then makes the gap between rice stock and total demand being larger year by year. This is such a big problem because Malaysia have to import the rice from country and affect its economic. For the boundary selection, this study will be focus on paddy production in Malaysia as a whole. The duration of this research will be 20 years from 2013 until 2033. This process is the same as the scope of the research part where it set the boundary selection of this study.

3.3.2 Formulation of Dynamic Hypothesis

Causal Loop Diagram (CLD) will describe the interactions and connections between the significant model variables depicted in the diagrams. The causal loop visualised how the variable obtained in problem articulation were related. It will show the connection using an arrow. It also has negative and positive links based on how the variable is connected with other variables. From the causal loop diagram, the base model can be developed.



From the figure above, we can see that the first positive loop with 'R' mark between birth and population. That population will increase the demand for the rice. The local production will then be divided by the local demand for rice between both variable. For the second bigger loop, it starts with paddy field that will increase the ratio to planting and harvest rice plant. Then, it will process to rice stock and make it as supply. The supply will be sold in the market and result in revenue as well as profit. The profit is again used to plant the rice and makes it a positive loop.

3.3.3 Formulation of Simulation Model

After the Causal loop has been visualised, a System Dynamic model can be developed. The Stock and Flow (SFD) diagram will be developed as a base model in this stage. It is an essential building block in system dynamic model. Stocks represent an accumulation of quantities, and Flow controls the changes to stocks via its rates. The basic base model will grow more extensive because of the related connection between stock, Flow and variable, making it a complete data model. This stage is the most crucial phase because the model development

occurs in this phase using Vensim software. This phase solved the full development of the model.

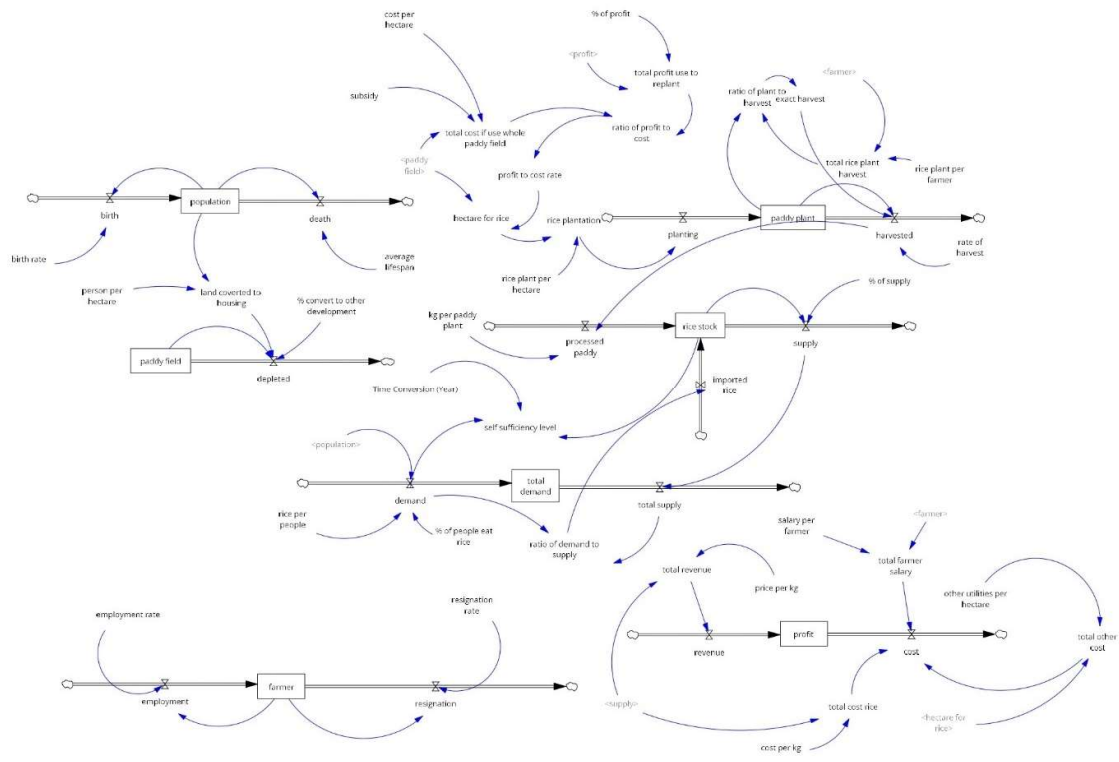


Figure 3 show the main model view, it start with population that need for development, development in other hand make the area of paddy field reduce. Paddy field use for planting rice plant. The population also will affect total demand and total supply. Rice stock will be divided by demand to get the self-sufficiency level.

3.4 Summary of Chapter

Methodology plays a vital role in writing research. It is because, in this chapter, we can see how the process of completing the model is made. The variables used in this model development can also be seen here, where the data and how it relates to each other will be obtained. SDM process consists of 5 processes has been explained briefly to make sure it is clear. The complete model cannot be done without following its written processes.

CHAPTER 4:

ANALYSIS AND FINDING

In this chapter, System Dynamic Research process will be continuing on the last process which is analysis including validation and policy implementation. It will start with the base case result, intervention, intervention result and drawback of the intervention itself.

4.1 Model Testing

4.1.1 Structure Validation Test

The first test in this process is structure validation, consisting of developing confidence that the structure of the developed model mimics the real-world structure pattern. There is an equation for every variable in the model. If the equation was put in correct way according to mathematical equation, there should not be a problem in the structure validation test.

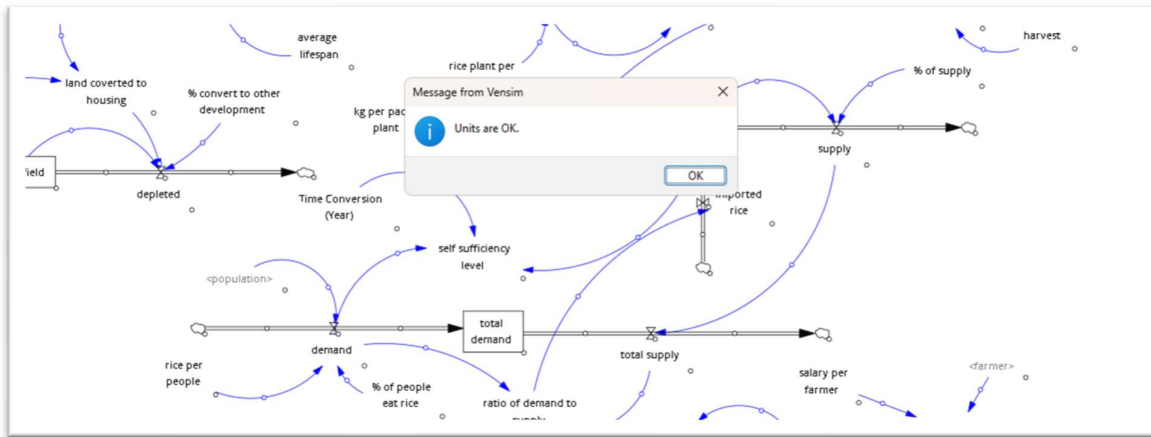


Figure 7 shows message box from Vensim that stated “Units are OK”. It means there are no error in equation of all the variables.

4.1.2 Behaviour Validation Test

Next, behaviour validation should be done after the structure validation test. It will be defined as developing confidence that the developed model's behaviour mimics the real-world behaviour pattern.

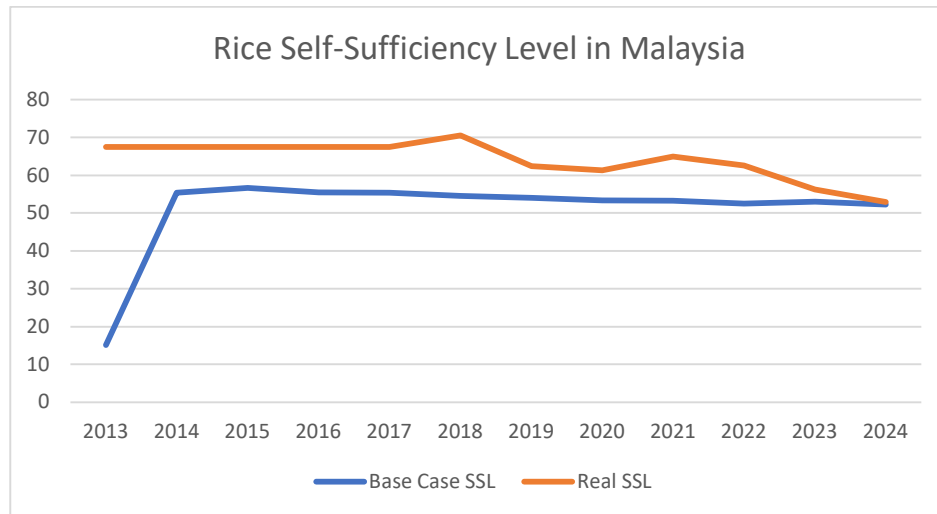


Figure 8 show comparison between real data and base case result of the rice self-sufficiency level in Malaysia. The self-sufficiency level is obtained by dividing the rice stock by the demand. This behaviour test is tested between 2013 until 2024. The calculation also had been done in Microsoft Excel as the table below.

Base Case SSL	Real SSL	Difference
15.1191	67.5	52.3809
55.3248	67.5	12.1752
56.6911	67.5	10.8089
55.5216	67.5	11.9784
55.3302	67.5	12.1698
54.5473	70.5	15.9527
53.9707	62.4	8.4293
53.3473	61.3	7.9527
53.3025	65	11.6975
52.4652	62.6	10.1348
53.0307	56.2	3.1693
52.2828	52.9	0.6172
Total Difference		157.4667
Average Difference		13.12223

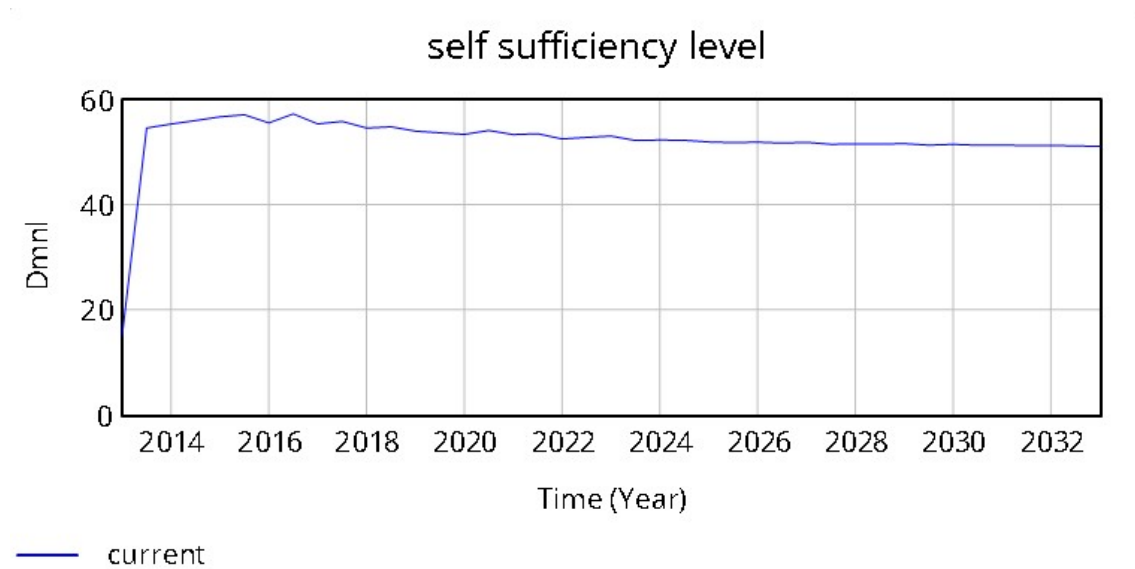
$$\begin{aligned}
 \text{Percentage Difference} &= \frac{\text{Average Difference}}{\text{Total Difference}} \times 100 \\
 &= \frac{157.4667}{13.12223} \times 100 \\
 &= 8.3333\%
 \end{aligned}$$

Table 3 show calculation between base case result and real data. When the average difference divide by the total difference and multiply by 100, the result is 8.3333%. Hence, we can conclude that the data is valid because the percentage difference is not more than 20%.

4.2 Policy Formulation and Evaluation

Optimization and best and worst-case scenarios are two crucial tools in this section. Changes in parameter values that take into account both the worst and best-case possibilities are referred to as best and worst-case scenarios. However, optimization necessitates the establishment of objective functions. The parameters that affect the objective function will then be selected. Lastly, the parameter that maximizes and minimizes the objective function is identified.

4.2.1 Base Case Result

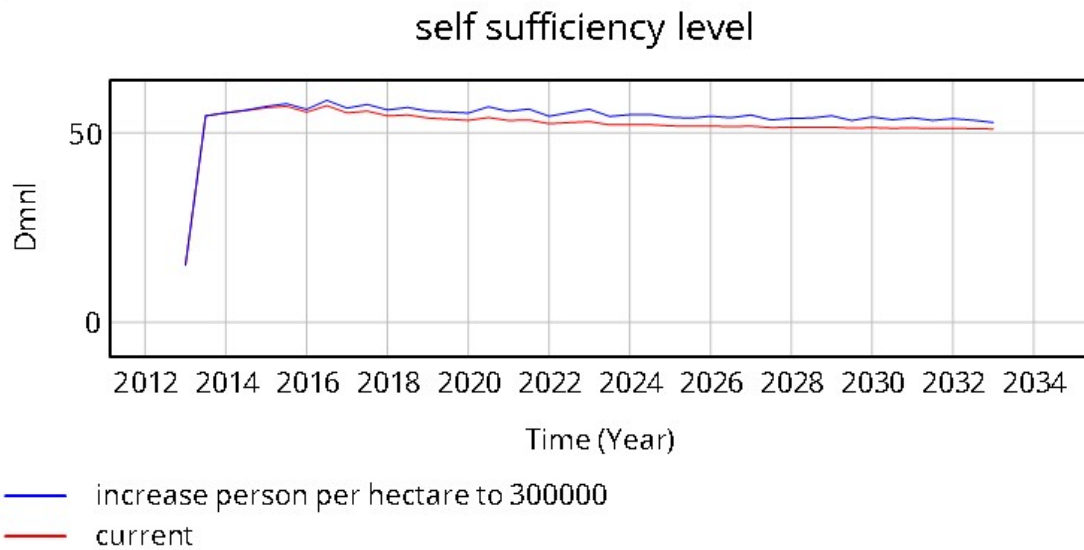


Graph above shows the rice self-sufficiency level in Malaysia as the years go by, influenced by some factors. Increasing of land that converts into development, lack of subsidies from the government and shortage of farmer are the factors that affect the rice stock, which directly affects the self-sufficiency level.

As shown on the figure above, the gap between rice stock and total demand were increasing once the system has been running. It not rapidly increases in 2012 to 2022 but still increasing

slowly and make the gap grow bigger as the year pass by. This scenario what they say is food security problem and government specifically Department of Agriculture should take an action to reduce those gaps.

4.2.2 Intervention 1 — Increase Person Per Hectare to 300,000



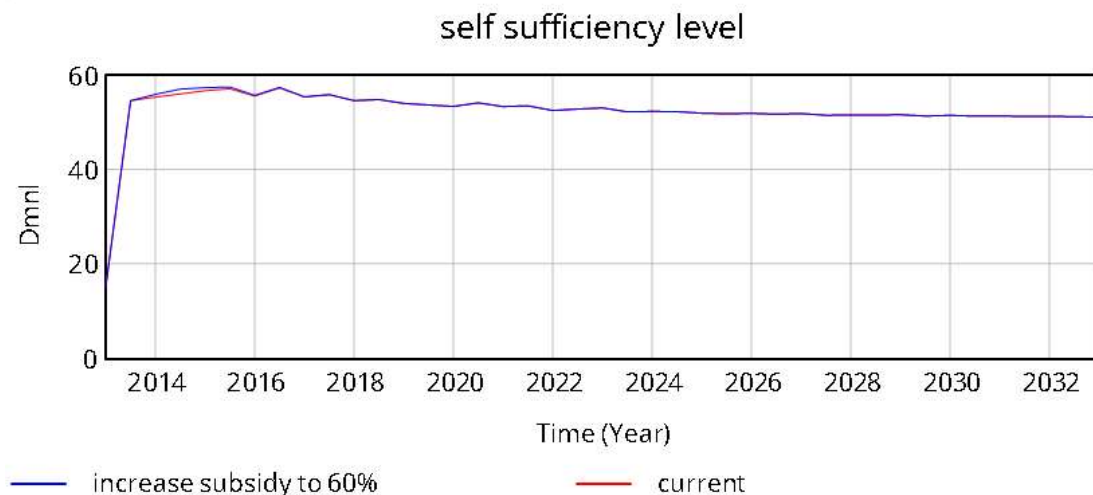
Year	Base Case	Intervention	Difference
2013	0.170757	0.170757	0
2013.5	54.8316	54.8316	0
2014	57.5418	57.5418	0
2014.5	57.7673	57.8606	0.0933
2015	58.6411	58.985	0.3439
2015.5	59.3335	60.0758	0.7423
2016	57.786	58.6451	0.8591
2016.5	59.7099	61.3075	1.5976
2017	57.6791	59.1475	1.4684
2017.5	58.1799	60.1833	2.0034
2018	56.7861	58.6138	1.8277
2018.5	57.0236	59.3146	2.291
2019	56.1112	58.2498	2.1386
2019.5	55.7257	57.9354	2.2097
2020	55.3716	57.6227	2.2511
2020.5	56.2031	59.4762	3.2731
2021	55.3359	58.161	2.8251
2021.5	55.4904	58.8192	3.3288
2022	54.3694	56.708	2.3386
2022.5	54.6882	57.7102	3.022
2023	54.9835	58.7481	3.7646
2023.5	54.0386	56.6379	2.5993
2024	54.1198	57.1179	2.9981
2024.5	54.0569	57.2092	3.1523
2025	53.6919	56.3561	2.6642

2025.5	53.534	56.0718	2.5378
2026	53.6309	56.6297	2.9988
2026.5	53.4466	56.1973	2.7507
2027	53.587	56.9773	3.3903
2027.5	53.1779	55.5465	2.3686
2028	53.2384	56.0076	2.7692
2028.5	53.2123	56.1024	2.8901
2029	53.3006	56.7657	3.4651
2029.5	52.9834	55.357	2.3736
2030	53.1304	56.3653	3.2349
2030.5	52.9544	55.5602	2.6058
2031	53.0136	56.1305	3.1169
2031.5	52.8652	55.3657	2.5005
2032	52.913	55.8922	2.9792
2032.5	52.8232	55.4295	2.6063
2033	52.7129	54.7327	2.0198
Total Base Case			2200.160657
Total Intervention			2292.560457

$$\begin{aligned}
 \text{Percentage Difference} &= \frac{(\text{Total Intervention} - \text{Total Base Case})}{\text{Total Base Case}} \times 100 \\
 &= \frac{(2292.560457 - 2200.160657)}{2200.160657} \times 100 \\
 &= 4.199684223\%
 \end{aligned}$$

The decreasing of conversion for development per people will reduce area that convert to other development. Hence, it will increase the paddy field area. Result from increase paddy field, rice stock will be increase and the self-sufficiency level of rice will be much better than the base case.

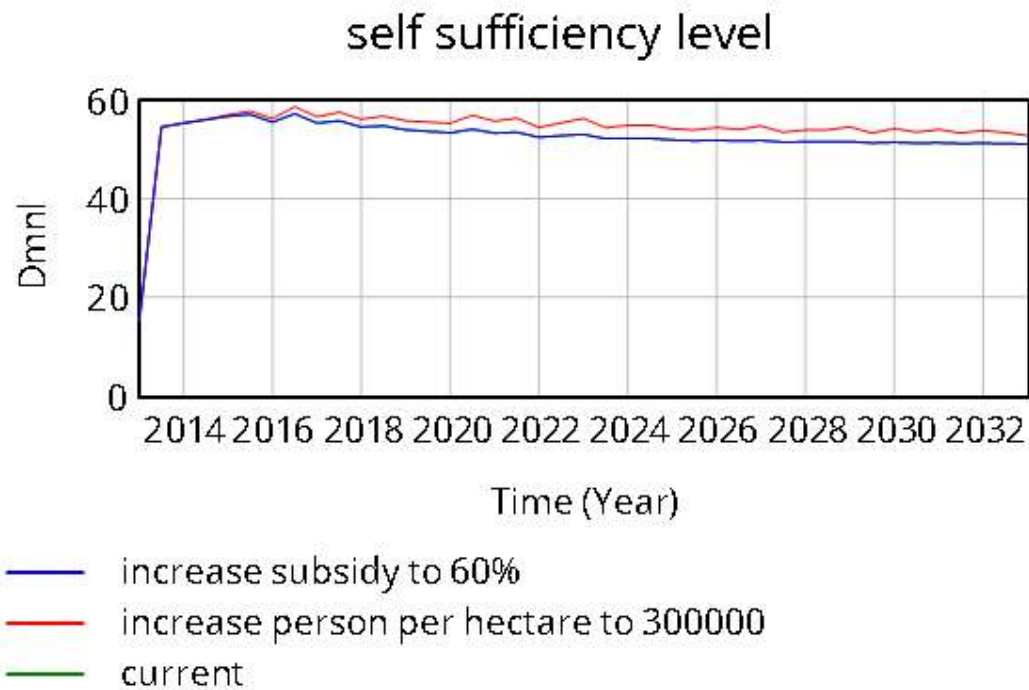
4.2.3 Intervention 2 — Increase Subsidy to 60%



Year	Base Case	Intervention	Difference
2013	0.170757	0.170757	0
2013.5	54.8316	54.8316	0
2014	57.5418	58.188	0.6462
2014.5	57.7673	58.4182	0.6509
2015	58.6411	59.228	0.5869
2015.5	59.3335	59.6931	0.3596
2016	57.786	57.9484	0.1624
2016.5	59.7099	59.8385	0.1286
2017	57.6791	57.7365	0.0574
2017.5	58.1799	58.2178	0.0379
2018	56.7861	56.8039	0.0178
2018.5	57.0236	57.0351	0.0115
2019	56.1112	56.1169	0.0057
2019.5	55.7257	55.7289	0.0032
2020	55.3716	55.3733	0.0017
2020.5	56.2031	56.2044	0.0013
2021	55.3359	55.3365	0.0006
2021.5	55.4904	55.4908	0.0004
2022	54.3694	54.3695	0.0001
2022.5	54.6882	54.6883	0.0001
2023	54.9835	54.9836	0.0001
2023.5	54.0386	54.0386	0
2024	54.1198	54.1199	0.0001
2024.5	54.0569	54.0569	0
2025	53.6919	53.6919	0
2025.5	53.534	53.534	0
2026	53.6309	53.6309	0
2026.5	53.4466	53.4466	0
2027	53.587	53.587	0
2027.5	53.1779	53.1779	0
2028	53.2384	53.2384	0
2028.5	53.2123	53.2123	0
2029	53.3006	53.3006	0
2029.5	52.9834	52.9834	0
2030	53.1304	53.1304	0
2030.5	52.9544	52.9544	0
2031	53.0136	53.0136	0
2031.5	52.8652	52.8652	0
2032	52.913	52.913	0
2032.5	52.8232	52.8232	0
2033	52.7129	52.7129	0
Total Base case			2200.160657
Total Intervention			2202.833157

$$\begin{aligned}
 \text{Percentage Difference} &= \frac{(\text{Total Intervention} - \text{Total Base Case})}{\text{Total Base Case}} \times 100 \\
 &= \frac{(2202.833157 - 2200.160657)}{2200.160657} \times 100 \\
 &= 0.121468402\%
 \end{aligned}$$

4.2.4 Intervention Result



From the figure, the rice self-sufficiency level for the intervention 1 is bigger than the intervention 2 which is add subsidy. Less land is converted to other uses, increasing the amount of paddy fields since there is less land available for other uses. Not only that, the percentage reduce for intervention 1 also slightly bigger, at 4.2% than intervention 2 which is just 0.12%. Thus, intervention 1 which is for increasing people per hectare will be chosen.

4.2.5 Possible Drawbacks from Intervention

With an increased people per hectare variable, this means that more land will be available to be converted into granary areas. However, increasing granary areas for rice production means that more labour will be required to carry out production in a granary area. With more labour required, this means labour costs will be driven higher.

4.3 Summary

Result for any study will be the most important part because reader will focus on that part. They want to know the finding for that study. In this study, the result was explained well with

a certain scenario to test it. In order for the reader to understand that the research that had been done also had the downside that the author had discovered, the drawback for the outcome had also been described.

CHAPTER 5:

CONCLUSION

5.1 Conclusion

The main objective of this study is to improve Malaysia's rice self-sufficiency level. Two interventions were carried out to achieve this objective. The first intervention was to increase people per hectare from 100,000 to 300,000. Doing so means that lesser land will be converted to housing projects, which leaves more land to be used as granary areas. This will directly increase the rice stock and directly increase the self-sufficiency level.

Next, the second intervention was to increase subsidy to 60%. Doing so means that the government will take up a cost to support the rice production, this covers all production costs that are relevant in the process. This directly supports more efficient production but having more efficient production does not increase rice stock by much.

This is evident because we can see that although minimal, the increase in self-sufficiency level when carrying out intervention 1 is higher than when carrying out intervention 2. This means intervention 1 is the ideal policy to carry out.

System Dynamics modelling provides decision-makers with a comprehensive understanding of how improvements in rice stock can enhance national food security. The simulation results reveal the relationships between key variables and their influence on rice stock levels. By examining these interconnected relationships, decision-makers can better understand how changes in one component of the system affect the overall performance of the rice industry. This enables the formulation of more informed and effective policies to achieve food security objectives. In this study, we have identified a few factors that affect the rice self-sufficiency level in Malaysia, a few being government subsidies, costs, percentage of land converted to development projects, and farmers. The problem can be solved after designing a policy for intervention, even if it means having to face some negative consequences.

5.2 Limitations

This study has several limitations that should be acknowledged. First, the accuracy of the System Dynamics model depends on the availability and quality of data used during model development. Although secondary data were obtained from government reports, journal articles, and other published sources, some variables lacked consistent or comprehensive historical data. As a result, several assumptions were made based on literature and expert judgement to complete the model.

Another limitation lies in the complexity of Malaysia's rice production system. The rice industry consists of numerous interconnected economic, environmental, and social factors, many of which are difficult to quantify or model accurately. Due to time constraints and the scope of an undergraduate research project, the model simplifies certain real-world processes and therefore may not fully represent the actual behaviour of the rice industry.

Furthermore, the interventions evaluated in this study were limited to selected policy variables identified from the literature. Other potentially influential factors, such as pest and disease outbreaks, extreme weather events, post-harvest losses, and variations in rice quality, were not incorporated into the model because of limited data availability and modelling constraints. Including these variables in future studies could improve the model's realism and predictive capability.

Despite these limitations, the developed System Dynamics model provides valuable insights into the relationships among key factors affecting Malaysia's rice production and self-sufficiency level. It also serves as a useful foundation for future research, where additional variables, more comprehensive datasets, and expert validation can further enhance the model's accuracy and policy relevance.

LIST OF REFERENCES

- Abdullah, M. Z. (2025). Impact of Climate Change on Paddy Productivity in Malaysia's Granary Areas: A Markov Chain Monte Carlo Analysis. *GATR Accounting and Finance Review*, 9(4), 01-12. doi:[https://doi.org/10.35609/afr.2025.9.4\(1\)](https://doi.org/10.35609/afr.2025.9.4(1))
- Abidin, I. S. (2022). The Role of Rural Infrastructure, Labour and Capital Investment on the Rice Production in Malaysia. *AGBIOFORUM*, 24(1).
- Ahmad Zairy, Z. A. (2023). *Sociological Issues and Challenges of Rice Production in Malaysia*. Food and Fertilizer Technology Center for the Asian and Pacific Region.
- Annuar, A., & Hisham, H. (14 April, 2025). *Malaysia faces uphill battle to boost rice self-sufficiency*. Retrieved from The Malaysian Reserve: <https://themalaysianreserve.com/2025/04/14/malaysia-faces-uphill-battle-to-boost-rice-self-sufficiency/>
- Baharudin, S. ' (2021). Understanding energy efficiency using a socioeconomic framework: The case of paddy farming in Malaysia. *Malaysian Journal of Society and Space*, 17(1).
- Çelik, M., & Serin, Z. V. (2022). A system dynamics approach to food security: The case of Turkey. *International Journal of Advanced and Applied Sciences*, 22-30.
- Dorairaj, D. &. (2023). Rice and Paddy Industry in Malaysia: Governance and policies, research trends, technology adoption and Resilience. *Frontiers in Sustainable Food System*, 7. doi:<https://doi.org/10.3389/fsufs.2023.1093605>
- Eneh, C., & Eneh, O. (2025). A system dynamics analysis of agricultural practices and food security in Nigeria. *Journal of Health, Population and Nutrition*, 44(1). doi:<https://doi.org/10.1186/s41043-025-01037-4>
- Firdaus, R. B. (2020). Paddy rice and food security in Malaysia: A review of climate change impacts. *Cogent Social Sciences*, 6(1). doi:<https://doi.org/10.1080/23311886.2020.1818373>
- Free Malaysia Today. (2025). *Only 10-15% of Malaysians buy local rice, says Mat Sabu*. Retrieved from Free Malaysia Today: <https://www.freemalaysiatoday.com/category/nation/2025/12/16/only-10-15-of-malaysians-buy-local-rice-says-mat-sabu>
- Fróna, D. (2020). Factors Affecting Food Security. *Annals of Faculty of Economics*, 1(1), 39-49.
- Khamko Thammavong, K. C. (2020). Analysis of Socio-Economic Factors Affecting Rice Production Systems Among Smallscale Farmers in Luang Prabang Province Laos. *International Journal of Sciences: Basic and Applied Research (IJSBAR)*, 54(2), 97-111.
- Khoo, G. S. (5 February, 2026). *Malaysia's rice self-sufficiency level stagnant at 52%, says Mat Sabu*. Retrieved from The Star: <https://www.thestar.com.my/news/nation/2026/02/05/malaysia039s-rice-self-sufficiency-level-stagnant-at-52-says-mat-sabu>

- Mgale, Y. J. (2020). Marketing efficiency and determinants of marketing channel choice by rice farmers in rural Tanzania: Evidence from Mbeya region, Tanzania. *Australian Journal of Agriculture and Resource Economics*, 64(4).
- Mills, E. (15 October, 2020). *Rice: How the world's staple is under pressure*. Retrieved from The World Economic Forum: <https://www.weforum.org/stories/food-water-air/how-much-is-a-bowl-of-rice-food-security-climate-change/>
- Poertner, H. R. (2023). Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. *Climate Change 2022 – Impacts, Adaptation and Vulnerability*, 719-906.
- Sokra, I. S. (2026). Factors Affecting Rice Production: A Systematic Review. *Journal of Agriculture and Technology*, 2(1), 19-46. doi:<https://doi.org/10.6084/m9.figshare.30814034>
- Solaymani, S. (2023). mpacts of Environmental Variables on Rice Production in Malaysia. *World*, 4(3). doi:<https://doi.org/10.3390/world4030028>
- Suffian, F. &. (2023). Policy Analysis: Food security in Malaysia. *Journal of Administrative Science*, 91-111.
- Suruhanjaya Persaingan Malaysia. (2024). *The Paddy And Rice Industry In Malaysia*. Kementerian Perdagangan Dalam Negeri Dan Kos Sara Hidup.
- Tan, B. T. (2021). Impact of Climate Change on Rice Yield in Malaysia: A Panel Data Analysis. *Agriculture*, 11(6). doi:<https://doi.org/10.3390/agriculture11060569>
- Tian, H. Z. (25 May, 2026). *Rice's Hidden Climate Cost: How Paddy Fields Are Driving Greenhouse Gas Emissions and What Can Be Done*. Retrieved from Down To Earth: <https://www.downtoearth.org.in/climate-change/rice-feeds-billions-of-people-but-its-role-in-fuelling-climate-change-is-growing>
- Zainudin, R. W. (2025). Sustaining Malaysia's Rice Economy: The Role of Economic Drivers in Food Security. *Jurnal Ekonomi Malaysia*, 59(2). doi:10.17576/JEM-2025-5902-2

APPENDIX

- 1) % convert to other development = 0.0005
Unit: 1/hectare/Year
- 2) % of people eat rice = 0.95
Unit: Dmnl
- 3) % of profit = 0.6
Unit: Dmnl
- 4) % of supply = 0.98
Unit: 1/Year
- 5) Average lifespan = 80
Unit: Years
- 6) Birth = population * birth rate
Unit: people/Year
- 7) Birth rate = 0.02
Unit: 1/Year
- 8) Cost = total cost rice + total farmer salary + total other cost
Unit: RM/Year
- 9) Cost per hectare = 800+STEP(850, 2026)+STEP(900,2028)+STEP(950,2030)
Unit: RM/hectare
- 10) Cost per Kg = 0.9
Unit: RM/Kg
- 11) Death = population / average lifespan
Unit: people/Year
- 12) Demand = Population * Rice per people * % of people eat rice
Unit: Kg/Year
- 13) Depleted = "% convert to other development"*land converted to housing*paddy field
Unit: hectare/Year
- 14) Employment = employment rate * farmer
Unit: farmer/Year
- 15) Employment Rate = 0.25
Unit: 1/Year

- 16) Exact Harvest = WITH LOOKUP(ratio of plant to harvest, LOOKUP([(0,0)-(100,10)],(0,1),(1,0.9),(100,0.1))
Unit: Dmnl
- 17) Farmer = INTEG(employment – resignation, 50000)
Unit: farmer
- 18) Harvested = Rate of harvest * Paddy plant * exact harvest
Unit: Plant/Year
- 19) Hectare for rice = paddy field*profit to cost rate
Unit: Hectare
- 20) Imported Rice = Ratio of Demand to supply
Unit: Kg/Year
- 21) KG Per paddy plant = RANDOM UNIFORM(0.03 , 0.05 , 1)
Unit: Kg/Plant
- 22) Land Converted to Housing = population * person per hectare
Unit: hectare
- 23) Other Utilities Per Hectare = 0.5
Unit: RM/Hectare/Year
- 24) Paddy Field = INTEG(- depleted, 90000)
Unit: hectare
- 25) Paddy plant = INTEG(planting – harvested, 90000*60*900)
Unit: plant
- 26) Person Per Hectare = 100,000
Unit: People/Hectare
- 27) Planting = rice plantation
Unit: plant/year
- 28) Population = INTEG(birth – death, 3.02e+07)
Unit: people
- 29) Price per Kg = 1.3
Unit: RM/Kg
- 30) Processed Paddy = harvested*kg per paddy plant
Unit: Kg/Year
- 31) Profit = INTEG(revenue – cost, 4e+07)
Unit: RM

- 32) Profit to cost rate = ratio of profit to cost
Unit: Dmnl
- 33) Rate of harvest = 0.9
Unit: 1/Year
- 34) Ratio of Demand to supply = IF THEN ELSE(demand>total supply, demand-total supply, 0)
Unit: Kg/Year
- 35) Ratio of plant to harvest = paddy plant/total rice plant harvest
Unit: Dmnl
- 36) Ratio of profit to cost = total profit use to replant/total cost if use whole paddy field
Unit: Dmnl
- 37) Resignation = resignation rate * farmer
Unit: farmer/Year
- 38) Resignation rate = 0.24
Unit: 1/Year
- 39) Revenue = total revenue
Unit: RM/Year
- 40) Rice per people = 80
Unit: Kg/People/Year
- 41) Rice plant per farmer = 366*1000
Unit: plant/farmer
- 42) Rice plant per hectare = 1000*70
Unit: Plant/hectare/Year
- 43) Rice Plantation = rice plant per hectare * hectare for rice
Unit: plant/Year
- 44) Rice Stock = INTEG(processed paddy + imported rice – supply, 2.63e+08)
Unit: Kg
- 45) Salary Per Farmer = 1200 + STEP(100 , 2020)+STEP(100, 2027)
Unit: RM/Farmer/Year
- 46) Self-sufficiency level = (rice stock/(demand*"Time Conversion (Year)"))*100
Unit: Dmnl
- 47) Subsidy = 1
Unit: 1/Year

- 48) Supply = Rice Stock * % of supply
Unit: Kg/Year
- 49) Time Conversion (Year) = 1
Unit: Year
- 50) Total cost if use whole paddy field = (cost per hectare*paddy field)*subsidy
Unit: RM
- 51) Total Cost Rice = Cost per kg * supply
Unit: RM/Year
- 52) Total Demand = INTEG(demand – total supply, 3.7e+08)
Unit: Kg
- 53) Total Farmer Salary = Salary per Farmer + Farmer
Unit: RM/Year
- 54) Total Other Cost = hectare for rice * other utilities per hectare
Unit: RM/Year
- 55) Total profit use to replant = profit * % of profit
Unit: RM
- 56) Total Revenue = Price per KG * supply
Unit: RM/Year
- 57) Total rice plant harvest = rice plant per farmer*farmer
Unit: plant
- 58) Total Supply = supply
Unit: Kg/Year